DSCI521 - Data Analysis and Interpretation
Final Presentation

User Spending and Churn Analysis

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Introduction:

Problem:

- Customer churn is a silent revenue killer. Understanding why users leave and predicting who might churn next can give businesses a competitive edge. By using transaction data, our objective is to uncover hidden patterns that signal potential churners before they leave.

Approach:

- Our approach involves
  - EDA on the Datasets
  - Pre-processing
  - Feature Engineering
  - Using various ML and DL models to determine the best approach to identify at-risk customers and optimize retention strategies.

Prior Works:

- Zadoo et al. (2022) - This study provides insights into using clustering techniques and feature selection for customer segmentation and churn prediction.
- Wu et al. (2021) - This paper integrates framework combining churn prediction and customer segmentation using techniques like Bayesian analysis and random forests.
- Nai et al. (2018) - The use of Singular Spectrum Analysis to model credit card usage behavior for different age groups provides a statistical approach to segmenting and analyzing demographic trends. This was used to understand factors that generally affect credit card usage among different demographics (especially age).
- Kara et al. (1996) - This is a very old empirical investigation of U.S. credit card users that studies the factors influencing card choice and usage behavior.

Dataset Introduction:

Dataset Source:

We used a synthetic dataset of users, their transaction and their card details from 2010-2019.

Link: https://www.kaggle.com/datasets/brtnsmth/intraday-market-data

Dataset Files:

- users_data.csv - Contains demographic and account-related details for each customer, serving as the foundation for user-specific attributes.
- transactions_data.csv - Logs all customer transactions, including amounts, timestamps, and merchant categories.
- cards_data.csv - Lists the payment methods associated with each user, their card details and their credit score related features.

Dataset Overview - Users Data:

Dimensions

- 2000 users x 14 features.

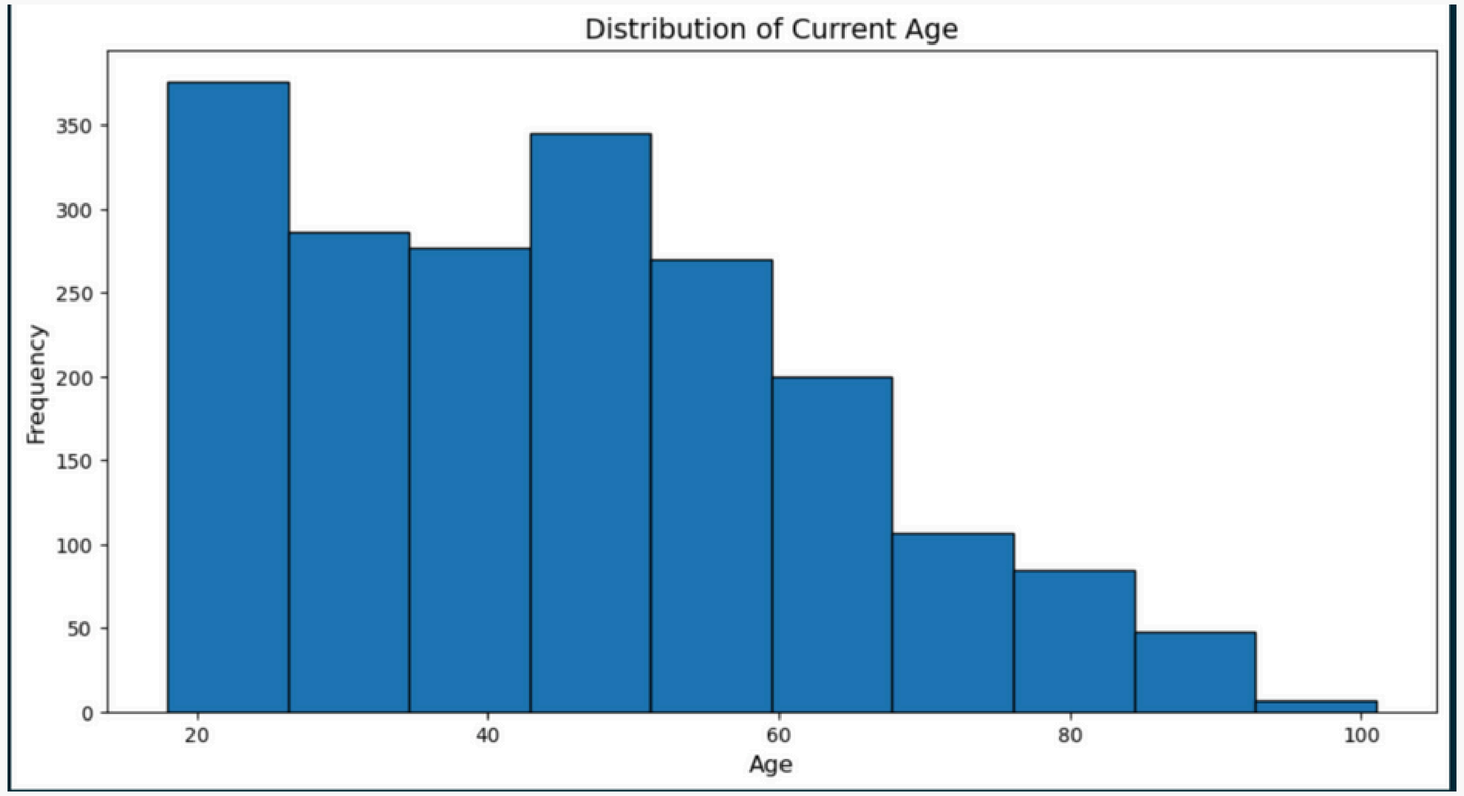
Major Variables:

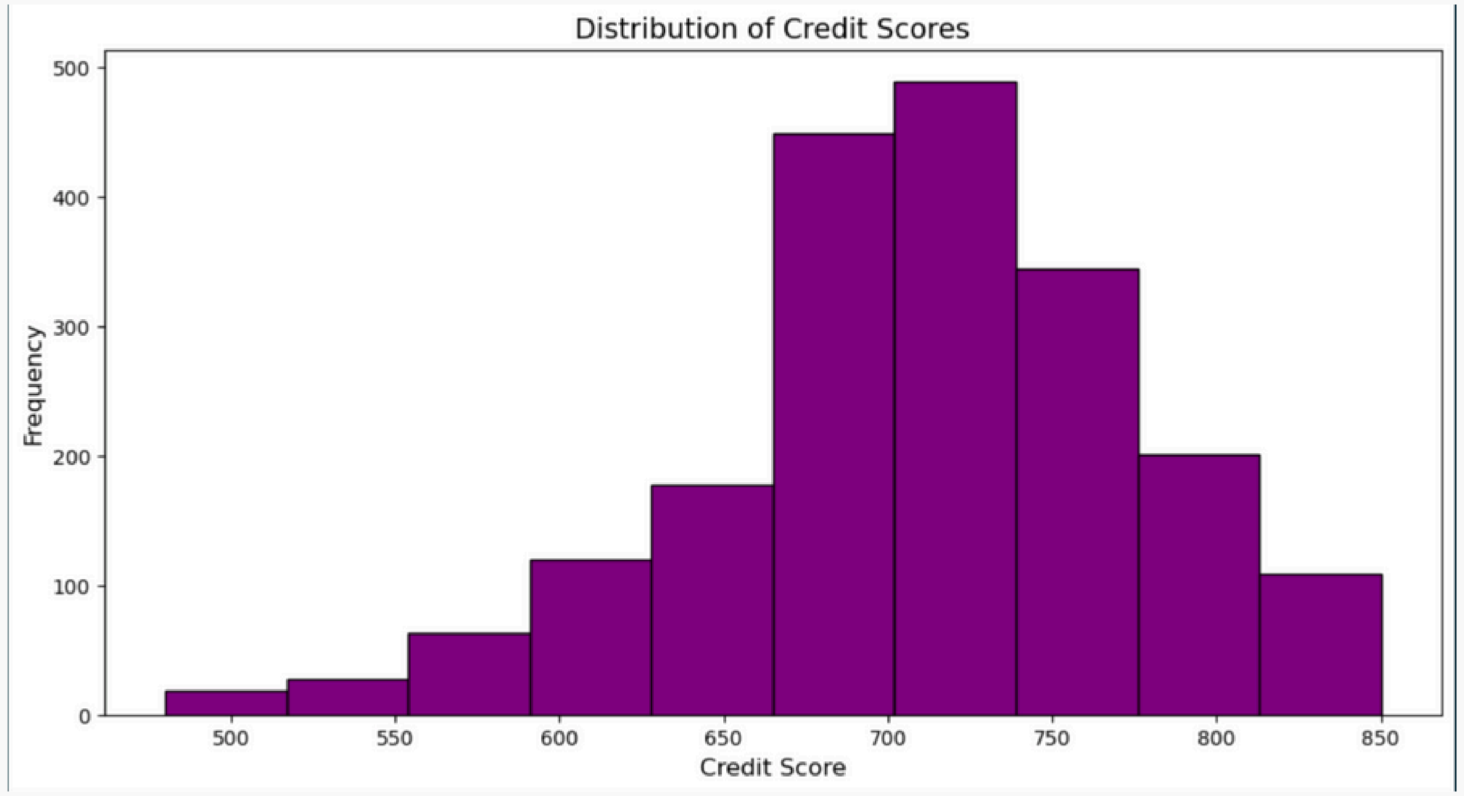
- id: Unique identifier for each user
- current_age, retirement_age, gender: User demographic information
- credit_score, total_debt, yearly_income: User financial information

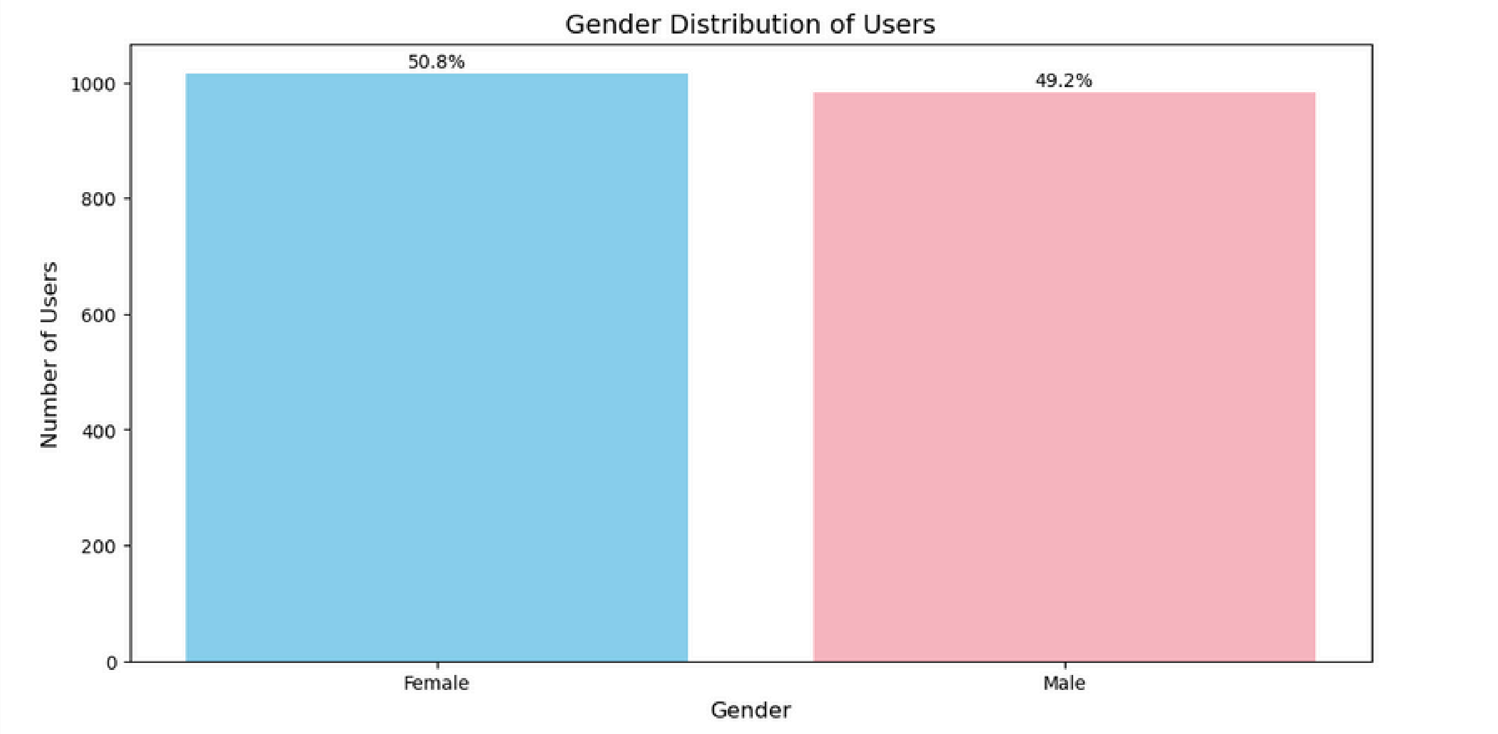
Variable Types:

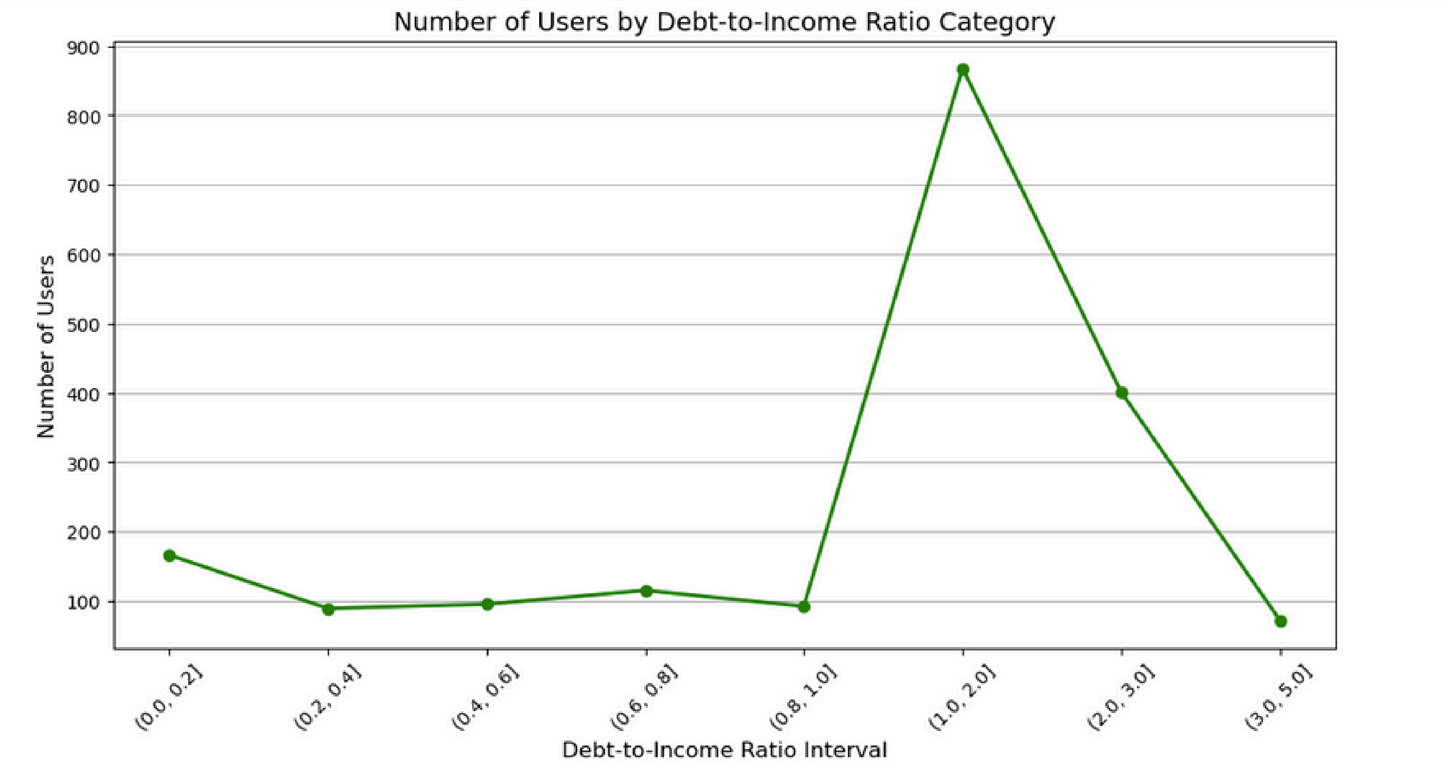
- Categorical: gender
- Numerical: current_age, credit_score, yearly_income

Dataset Overview - Users Data:

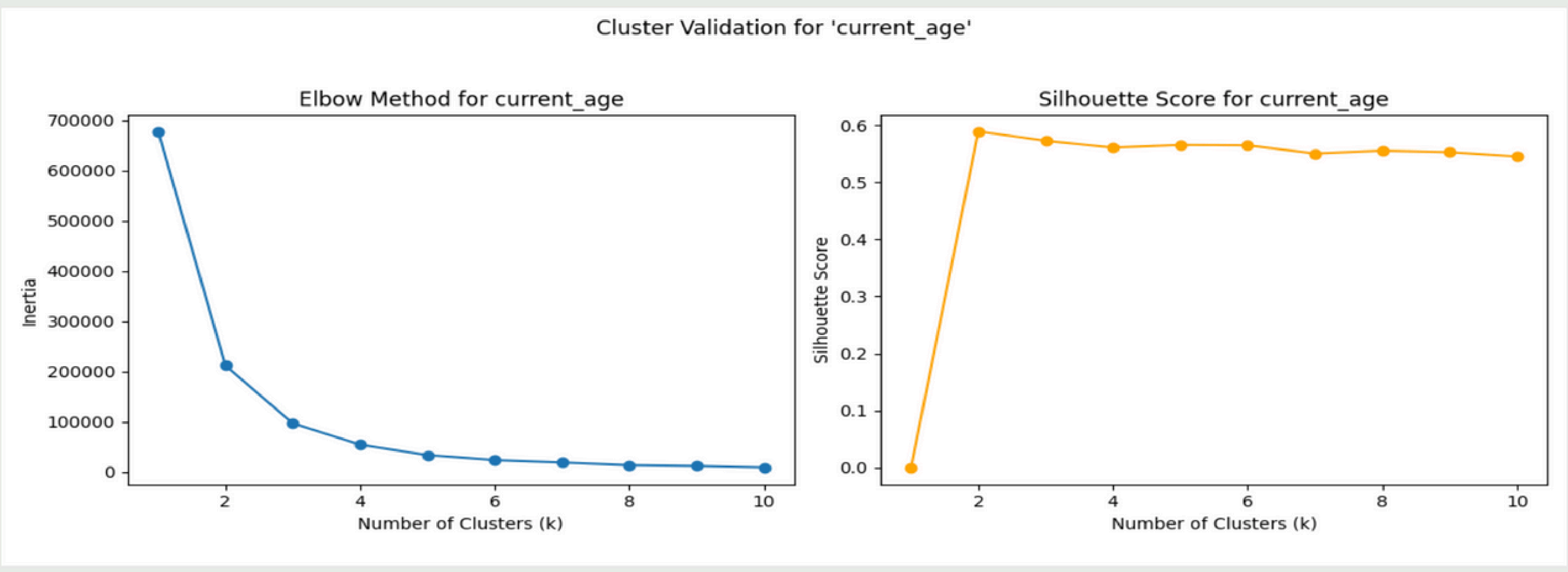


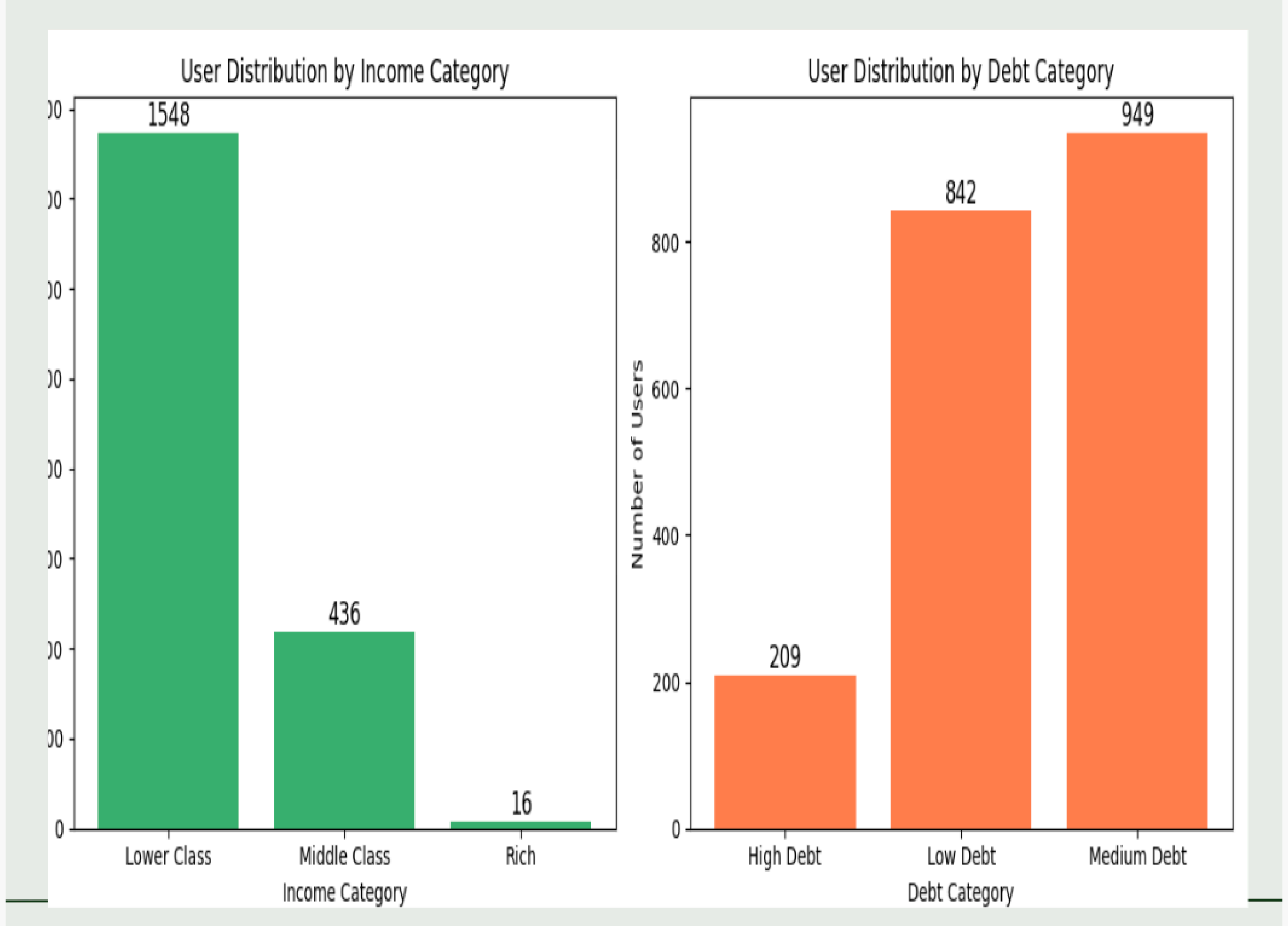






Dataset Overview - Users Data:





Dataset Overview - Transactions Data:

Dimensions

- 14 mil transactions x 12 features.

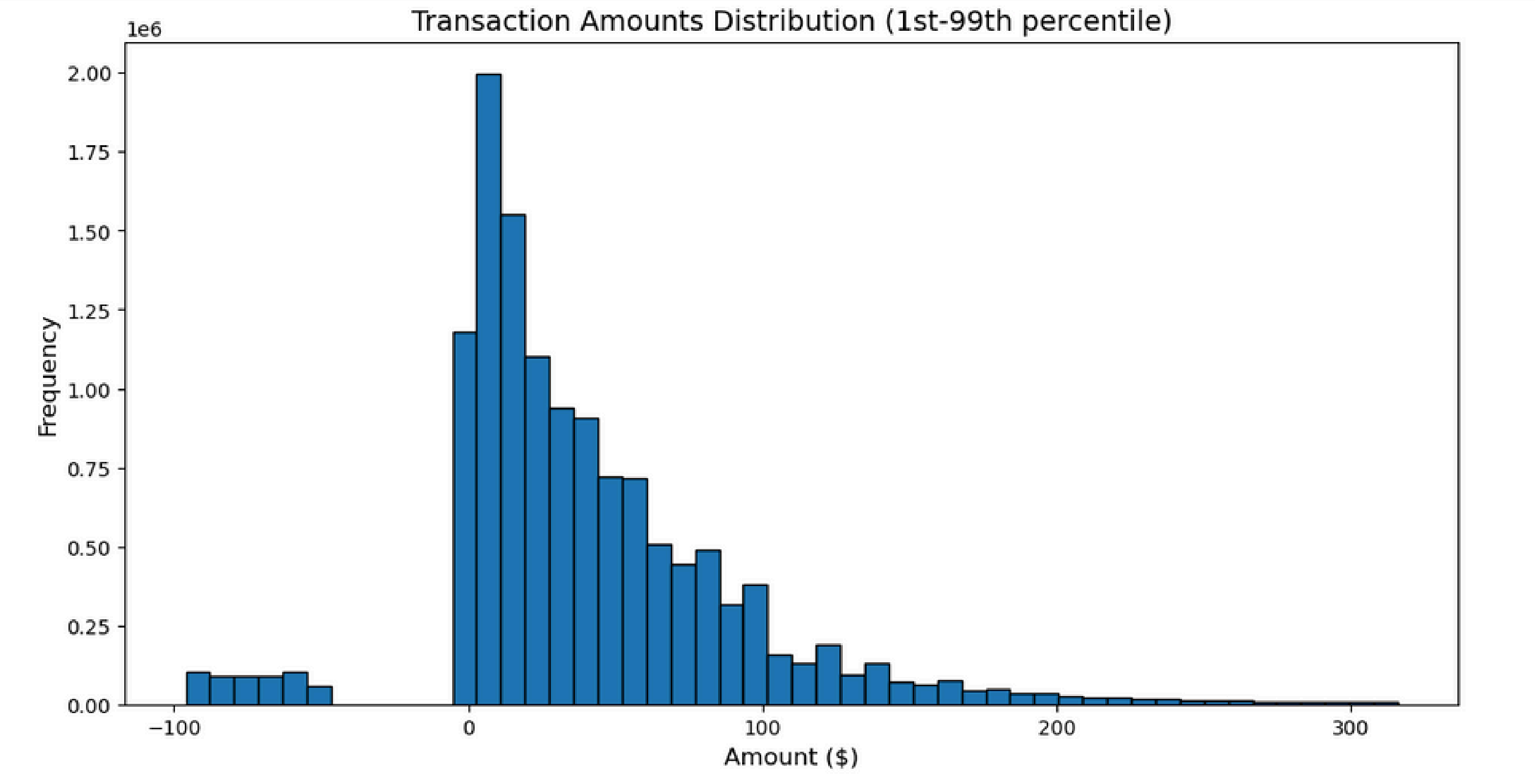
Major Variables:

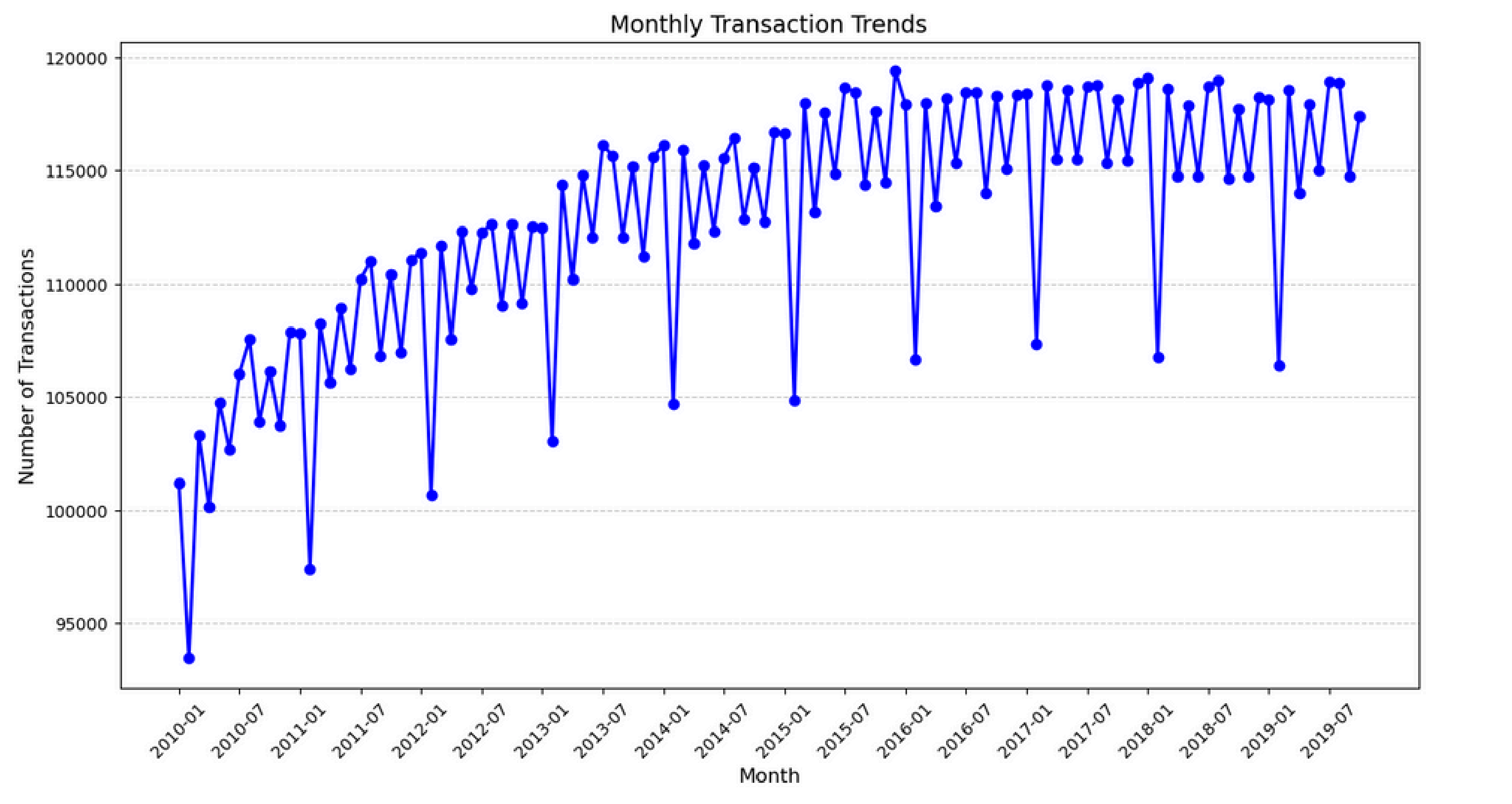
- id: Unique identifier for each transaction
- date, client_id, card_id: Transaction specific information
- merchant_id, merchant_city: Merchant demographic information
- use_chip, mcc: Transaction method and type

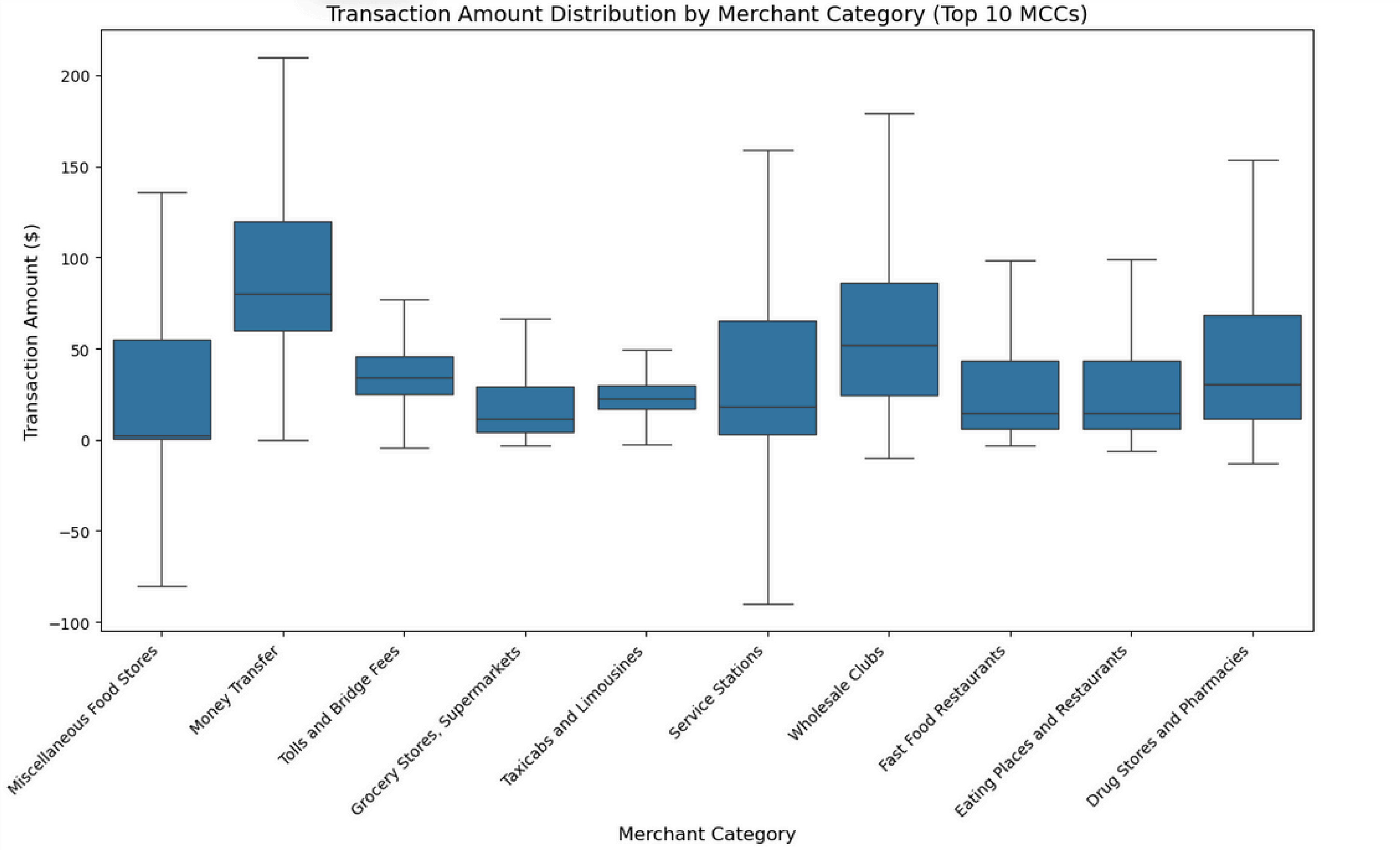
Variable Types:

- Categorical: use_chip, merchant_city, merchant_state
- Numerical: amount, card, mcc
- Date/Time: date

Dataset Overview - Transactions Data:







Dataset Overview - Cards Data:

Dimensions

- 6146 x 13 features.

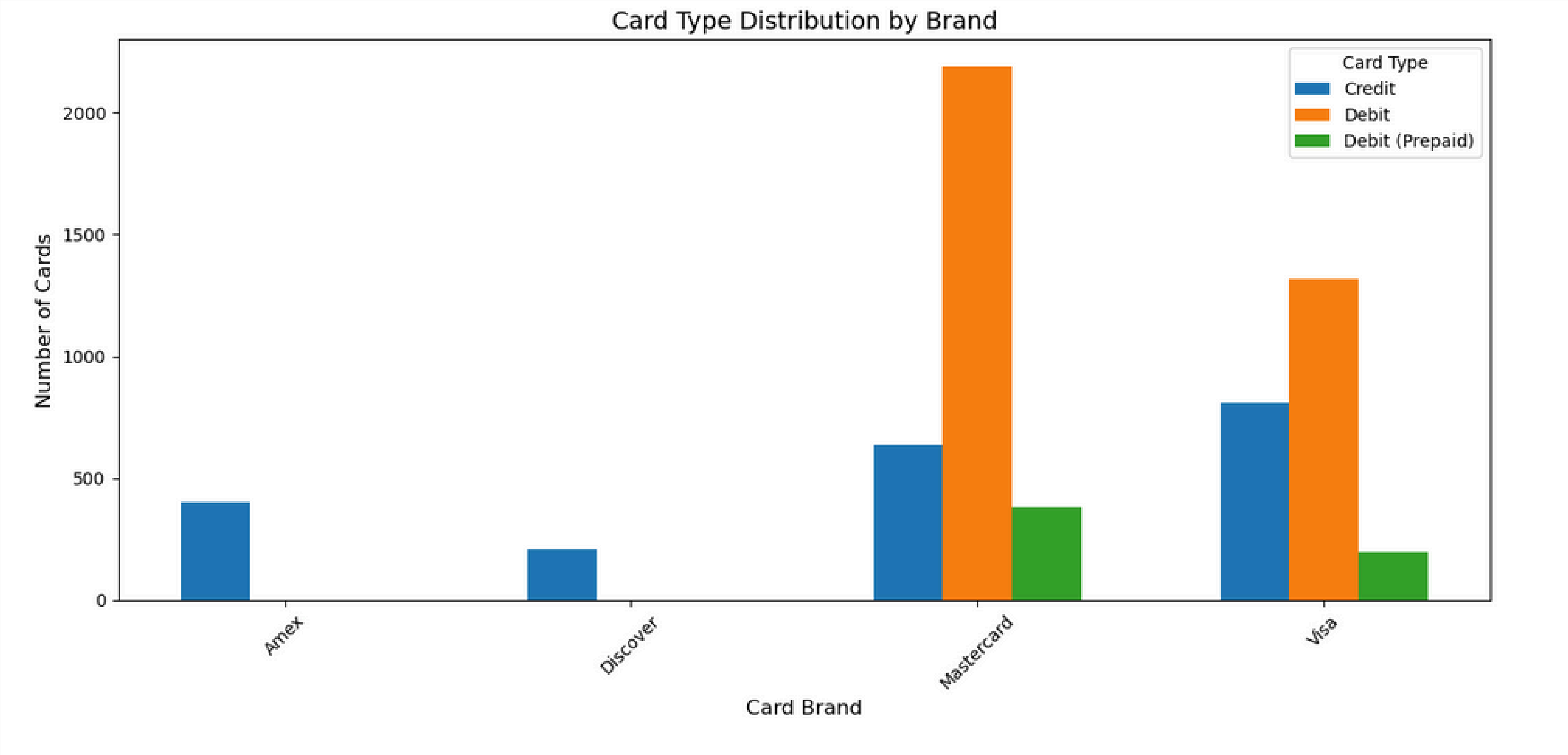
Major Variables:

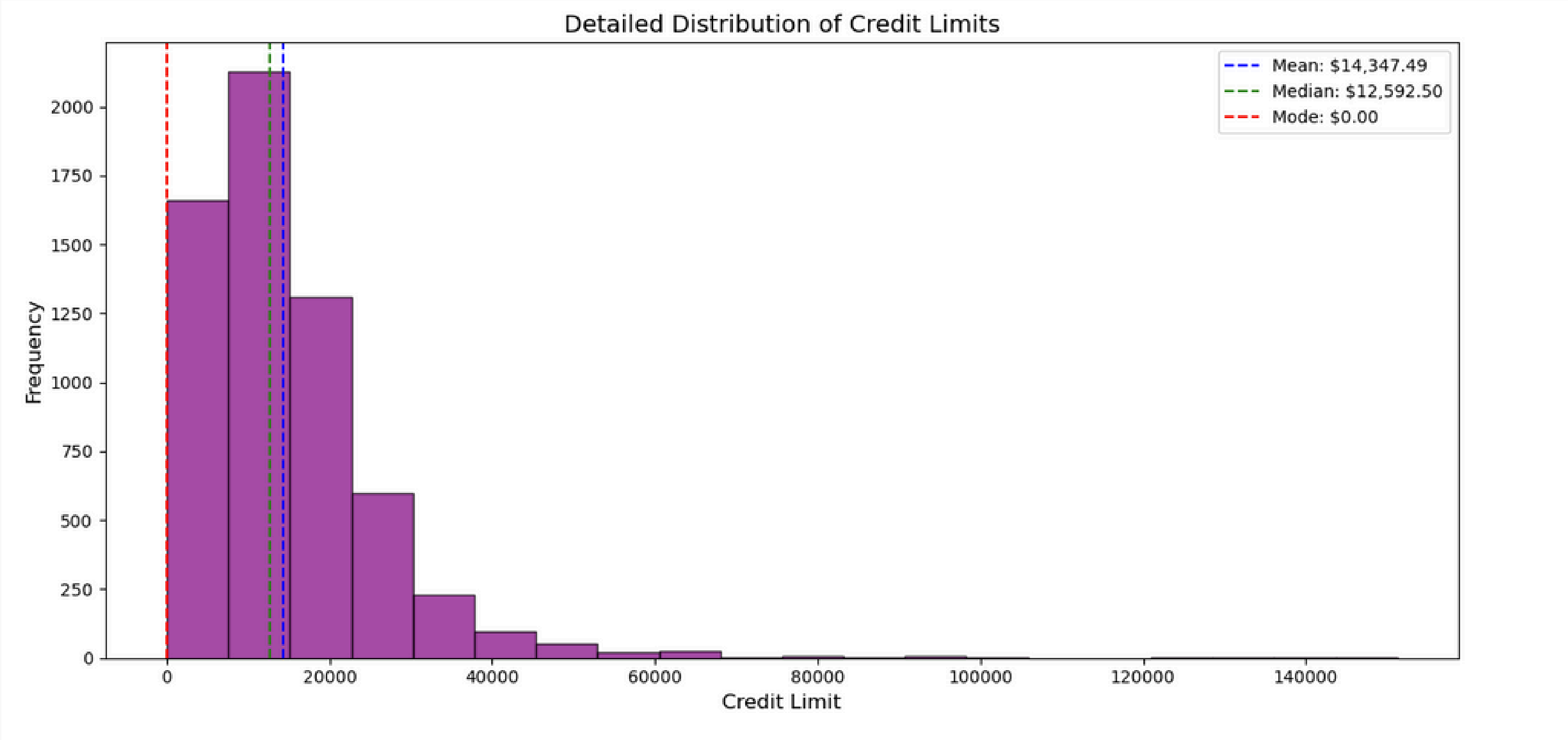
- id, card_number, expires: Card details
- client_id, acct_open_date, credit_limit: User and account information
- card_brand, card_type, has_chip: Card types and security

Variable Types:

- Categorical: card_brand, card_type, has_chip, card_on_dark_web
- Numerical: num_cards_issued, cvv, year_pin_last_changed, client_id
- Date/Time: expires, acct_open_date

Dataset Overview - Cards Data:





Common IDs for Datasets:

Common ID for Merging

- client_id - Key column that links all the datasets together.
- merchant_id - Key column for understanding relationship between users, transactions and individual merchants.

Feature Eng.: Spending Behavior

New features created:

- **Total Spending** - The sum of all transaction amounts made by a user, representing their overall expenditure.
- **Total refunds** - The total amount refunded to a user, indicating potential dissatisfaction or product returns.
- **Average and Median transaction amounts** - The mean and median of a user's transactions, capturing their typical spending behavior.
- **Max and Min transaction amounts** - The highest and lowest transaction amounts, highlighting spending extremes.
- **Spending Variability** - The standard deviation of transaction amounts, measuring fluctuations in a user's spending habits.

Feature Eng.: Transaction Frequency

New features created:

- **Total Transactions -** The total number of transactions made by a user, representing their engagement level.
- **Months Active -** The duration (in months) between a user's first and last transaction, indicating their activity span.
- **Transactions Per Month -** The average number of transactions per month, showing how frequently a user makes purchases.
- **Days Since Last Transaction -** The number of days since a user's most recent transaction, highlighting recency of activity.

Feature Eng.: Payment Method

New features created:

- **Most Frequent Payment Method** – The payment method (Online, Card, or Swipe) that a user uses most often, indicating their preferred transaction type.
- **Total Spending per Method** – The total amount spent by a user across different payment methods, helping understand their spending distribution.
- **Total Transactions per Method** – The count of transactions per payment method, showing how often each method is used.
- **Preferred Payment Method** – The payment method with the highest total spending, highlighting where a user spends the most money.

Feature Eng.: Merchant Behavior

New features created:

- **Most Frequent Merchant Category –** The most common merchant category a user transacts with, determined using the Merchant Category Code (MCC), indicating their preferred type of business.
- **Average Transaction Per Merchant –** The average spending per merchant by a user, calculated by dividing total spending by the number of unique merchants they have interacted with.

Feature Eng.: Churn Label

Target features creation methodology:

1. Calculated the average number of transactions per month for each user.
2. Split transactions into historical (before last 6 months) and recent (last 6 months) periods.
3. Compared the recent transaction rate to the historical average.
4. If recent activity dropped below a set threshold, the user was labeled as churned (1); otherwise, as active (0).
5. Users with no transactions in the last 6 months were automatically classified as churned.

Churn Label distribution

1. 64% active users vs 34% Churned Users

Feature Eng.: Preprocessing

Step 1: Filtering Columns

1. Removed variables that were directly used for calcualtion of our label or were redundant to our model.
2. Eg: client_id, days_since_last_transaction, transactions_per_month, historical_avg_transactions

Step 2: Removing Multicollinearity:

1. Set a threshold of 0.8 to remove all variables that were multicollinear.
2. per_capita_income and yearly_income
3. total_transactions and transactions by other methods etc.

Step 3: Split into Training and Test data:

1. Used a 80-20 split for seperating the training and testing data

Feature Eng.: Preprocessing

Step 4: Standardizing Numerical Variables:

1. Standardized all our numerical variables using standard z-scoring method.

Step 5: One-hot-encoding Simple Categorical Variables:

1. One-hot encoded the simple categorical variables in our dataset like: gender, most_frequent_merchant_category, most_frequent_payment_method, preferred_payment_method

Step 6: Frequency encoding of High Cardinality Categorical Variables:

1. This function applies frequency encoding to the most_frequent_merchant column by replacing each merchant with its occurrence count in the training data. Missing values in the test data are filled with 0 to handle unseen merchants.

Models: Logistic Regression

Why?

We wanted to use Logistic Regression as our starting model because of it’s simplicity. Using this as a base model to estimate the probability of the churn, we wanted to move ahead.

Model Definition

- A very simple sigmoid function.
- 5000 max iterations
- C = 0.5 (which is used to make sure our model does not overfit)

Models: Logistic Regression

Result metrics:

- Accuracy: Training 76.06% || 74.00% Test
- Precision: Training 91.97% || 89.95% Test
- Recall: Trainning 68.46% || 66.7% Test
- F1 Score: Training 78.50% || 76.58% Test
- ROC AUC: Training 85.26% || 82.86% Test

Important Features:

- Preferred payment method - Online
- Tolls and Bridge fees category
- Drinking Places (Alcohol beverage) category
- Spending variability
- Male

Models: Complex Logistic Regression

Model Definition:

- Variable values of C
- L1 and L2 regularization
- Different solvers
- Cross-Validation
- Class Weight

Result metrics:

- **Accuracy: Training 67.56% || 65.00% Test**
- **Precision: Training 70.78% || 69.36% Test**
- **Recall: Trainning 83.74% || 80.78% Test**
- **F1 Score: Training 76.72% || 76.78% Test**
- **ROC AUC: Training 77.42% || 77.42% Test**

Models: XG Boost Modelling

Why?

We used XGBoost because it is a powerful gradient boosting algorithm that improves churn prediction by capturing complex patterns in data. It can also handle large datasets, reduces overfitting through regularization, and allows usage of several different underlying algorithms.

Model Definition

- LogLoss eval metric
- Lambda and Alpha regularization
- Hyperparameters: Learning rate, max depth, subsampling
- Early Stopping

Models: XG Boost Modelling

Result metrics:

- Accuracy: Training 76.62% || 73.00% Test
- Precision: Training 82.50% || 81.01% Test
- Recall: Trainning 76.69% || 75.29% Test
- F1 Score: Training 80.72% || 78.05% Test
- ROC AUC: Training 85.56% || 82.20% Test

Important Features:

- Transportation Category
- Total spent by chip transactions
- Refunds
- Credit score
- Spending variability

Models: Deep Learning Model

Why?

Deep Learning allows us to use multi-layer neural networks to capture complex relationships in data. It can identify intricate patterns that traditional models may miss, making it useful for churn prediction in our case.

Model Definition

- 5 Layers: 256 -> 128 -> 64 -> 32 -> 1
- Activation Functions: LeakyReLu and ReLu with Sigmoid and final classifier
- Adam Optimizer
- BCE with Log Loss function
- Hyperparameters: Dropout rate, learning rate
- 500 epochs

Models: Deep Learning Model

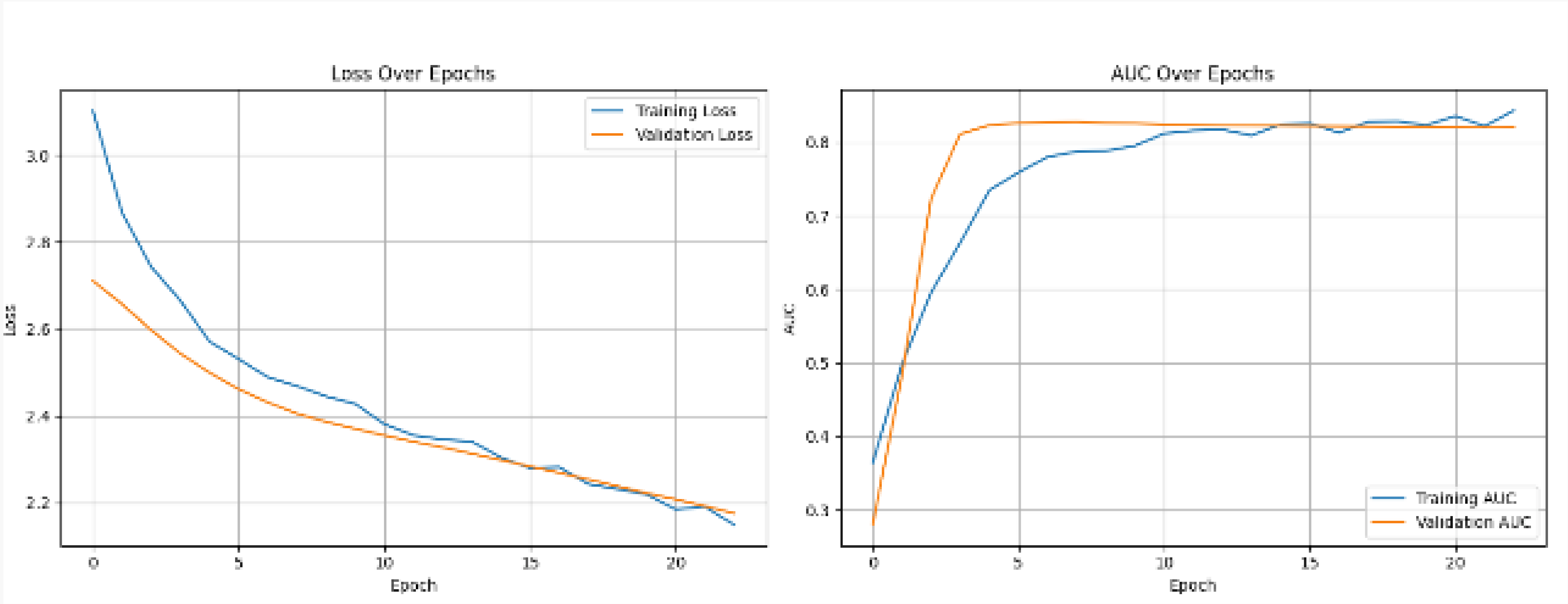
Result metrics:

- Accuracy: 76.00%
- Precision: 100.00%
- Recall: 62.35%
- F1 Score: 76.81%
- ROC AUC: 82.79%

Confusion Matrix:

- TP: 145
- FP: 0
- TN: 96
- FN: 159

Models: Deep Learning Model



Models: Honorable Mentions

SVM:

We tried using Support Vector Machines for our modeling but unfortunately due to the vast number of features, the runtime of SVM was not computational for practical purposes.

Using PCA with each Model:

Even using PCA before each model and removing about 10 features (retaining 90%) variance did not help us with better metrics.

Final Results

Best performance in each category:

- Accuracy - XG Boost (Train: 76.62%, Test: 73.00%)
- Precision - Deep Learning (Test: 100.00%)
- Recall: XGBoost (Train: 76.69%, Test: 75.29%)
- F1 Score: XGBoost (Train: 80.72%, Test: 78.05%)
- ROC AUC: Logistic Regression (Train: 85.26%, Test: 82.86%)

Most common features

- Demographics: Age, Gender
- Payment method: Online transactions
- Categories: Recreational, Drinking Places, Transportation and Gas Fees, Beauty and Pharma.
- Financial: total_refunds, credit_score

Areas for Improvement

Advanced Feature Engineering:

- Extracting more sophisticated behavioral patterns, such as time-based spending trends or clustering users based on transaction habits, could improve model performance.

RNN-Based Model:

- Since transactions occur over a long time, a Recurrent Neural Network (RNN) or LSTM could better capture sequential dependencies and provide a more dynamic churn prediction approach.

Real-World Dataset:

- Applying this model to an actual business dataset would give more realistic insights, helping fine-tune features and improve generalization.

References

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Thank you